

IoT Summer School 2026

Theory Assignment

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Q9. Explain the difference between git pull, git fetch, and git clone. When would you use each?

git clone

- Copies the entire remote repository to your local computer.
- **Use:** When downloading a repository for the first time.

git fetch

- Downloads the latest changes from the remote repository without merging them into your local branch.
- **Use:** When you want to check updates before merging.

git pull

- Downloads the latest changes and automatically merges them into your local branch.
- **Use:** When you want to update your local repository with the latest changes.

Q10. What is a .gitignore file? Write a .gitignore for an Arduino project.

A **.gitignore** file tells Git which files and folders should not be tracked or uploaded to the repository.

Example .gitignore

```
*.hex  
*.elf  
.DS_Store  
Thumbs.db  
.vscode/  
build/
```

Q19. What is the difference between analogWrite() and analogRead() in Arduino? What is PWM and why is it used? Give a practical IoT example for each.

Difference

analogRead()	analogWrite()
Reads analog input values.	Generates PWM output.
Returns values from 0–1023.	Accepts values from 0–255.
Used with analog sensors.	Used to control brightness or speed.

What is PWM?

PWM (Pulse Width Modulation) is a technique that simulates analog output by rapidly switching a digital signal ON and OFF.

Why is PWM used?

PWM is used to control LED brightness, motor speed, and power efficiently.

Practical IoT Examples

- **analogRead()**: Reading an LDR sensor in a smart street light.
- **analogWrite()**: Controlling LED brightness based on ambient light.

Q20. Explain the **setup()** and **loop()** functions in Arduino. What happens if you put a long **delay()** inside **loop()**? How does this affect sensor reading responsiveness? What is the non-blocking alternative?

setup()

- Runs only once when the Arduino starts or resets.
- Used to initialize pins, Serial communication, and libraries.

loop()

- Runs continuously after **setup()**.
- Contains the main program logic.

Effect of long delay()

A long **delay()** pauses program execution. During this time the Arduino cannot respond to sensors or button presses, making the system less responsive.

Non-blocking alternative

Use the **millis()** function instead of **delay()** so the Arduino can perform multiple tasks without stopping.

Q29. What is sensor calibration and why is it important in IoT systems? How would you calibrate an analog soil moisture sensor to give accurate percentage readings? Describe the two-point calibration method.

Sensor Calibration

Sensor calibration is the process of adjusting a sensor so that it provides accurate and reliable measurements.

Importance

- Improves accuracy.
- Reduces measurement errors.
- Ensures reliable IoT data.

Two-Point Calibration Method

1. Measure the sensor value in **completely dry soil** and record it as the dry value.
2. Measure the sensor value in **fully wet soil** and record it as the wet value.
3. Map all future readings between the dry and wet values to calculate the soil moisture percentage (0%–100%).

Q30. Explain I2C protocol with respect to IoT sensor communication. What are SDA and SCL lines? How does I2C addressing work? Name 3 common IoT sensors that use I2C.

I2C Protocol

I2C (Inter-Integrated Circuit) is a communication protocol that allows multiple sensors and devices to communicate with a microcontroller using only two wires.

SDA and SCL

- **SDA (Serial Data):** Transfers data between devices.

- **SCL (Serial Clock):** Synchronizes data transfer using clock signals.

I2C Addressing

Each I2C device has a unique address. The master (Arduino) communicates with a specific device by sending its address before data transfer.

Common IoT Sensors Using I2C

- **BMP280** (Temperature and Pressure Sensor)
- **MPU6050** (Accelerometer and Gyroscope)
- **BH1750** (Light Intensity Sensor)